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Institute of Management & Career Courses (IMCC)

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DEPARTMENT OF COMPUTER APPLICATIONS (MCA)



CAPSTONE PROJECT SCOPE PROPOSAL & TECHNICAL SYNOPSIS

Academic Cohort: 2026-2028 • Class: FY MCA • Section: Division A • Group 17

REVISION REQUESTED

Project Title	VIVA – Voice Interactive Virtual Interview Assistant		
Team Name	Twin_Stack Group 17	Project Track / Domain	Web & Mobile Full Stack Application
Project Category	Capstone Team Project	Submission Date	6 October 2026
Internal Faculty Guide	Ms. Kalpana Dhende ksd.imcc@mespune.in	External Industry Guide	AUTO Industry Organization

1. STUDENT PROJECT TEAM ROSTER

TOTAL MEMBERS: 2

#	STUDENT FULL NAME	ROLL NO. / PRN	INSTITUTIONAL EMAIL	PROJECT RESPONSIBILITY
1	Tanvi T. Joshi	2601086	tanvitjoshi20@gmail.com	Lead Team Leader (Lead)
2	Mayuri R. Gandhi	2601062	mayurigandhi005@gmail.com	Core Team Member

2. PROBLEM STATEMENT & ACADEMIC SYNOPSIS

Students prepare for interviews with theory questions and text-based mock tests, but real interviews are spoken. Because they get no speaking practice, many students feel nervous and cannot explain their answers clearly. Existing mock test websites use fixed questions, ignore the student's resume, and give only a score without personal feedback.

3. PROJECT SCOPE & FUNCTIONAL OBJECTIVES

1. To give students a real interview-like experience using voice-based questions and answers, with no camera needed.
2. To generate questions from the student's resume.
3. To ask follow-up questions based on the student's previous answers.
4. To allow both Voice Mode and Text Mode, with Easy, Moderate and Hard levels.
5. To provide human evaluation with marks and feedback.
6. To show scores, history and progress on a dashboard.

4. TECHNOLOGY STACK & INFRASTRUCTURE ENGINES

PRIMARY DEVELOPMENT & ARCHITECTURAL STACK	DATABASES, CACHES & STORAGE ENGINES
Python Django HTML5 CSS3 JavaScript	SQLite

5. ARCHITECTURAL MODULE BREAKDOWN & DELIVERABLES

8 DEFINED MODULES

#	FUNCTIONAL MODULE	DETAILED SYSTEM SPECIFICATION & DELIVERABLE SCOPE
M1	Module 1: User Registration and Login	Allows students, evaluators and admin to register and log in with their own role. Input: name, email, password and role. Passwords are stored securely, and each user can see only their own data.
M2	Module 2: Resume Scanner	The student uploads a resume in PDF format. The system reads the text and extracts education, skills, projects, certifications and experience, and saves them for resume-based questions.
M3	Module 3: Question Bank	Stores interview questions by type (HR, Technical, Resume-Based, General) and by level (Easy, Moderate, Hard). The admin can add, edit and delete questions.
M4	Module 4: Voice-Based Virtual Interviewer	Asks questions in voice using text-to-speech and records the student's spoken answer. Speech-to-text converts the answer into text, which is shown on screen and saved in the transcript.
M5	Module 5: Dynamic Follow-Up Questions	Reads the student's previous answer, finds keywords and asks a related next question, for example "Why did you choose SQLite for your project?". It does not repeat questions and follows the selected level.
M6	Module 6: Voice and Text Modes	The student can answer by speaking (Voice Mode) or by typing (Text Mode). The full transcript is saved in both modes. Audio is not stored, only the text.
M7	Module 7: Human-Based Evaluation	The evaluator reviews the interview transcript and gives marks and feedback on correctness, technical knowledge, communication, grammar, clarity and response time. The marks and feedback are saved in the database.
M8	Module 8: Dashboard and Report	Shows the student's overall score, question-wise marks, evaluator feedback, improvement areas, interview history and progress. It also generates the final interview report as a PDF.

6. TANGIBLE DELIVERABLES & EXPECTED OUTPUT

The system will give students an interactive and personalized interview preparation environment. Students can practice answering questions by speaking, improve their communication skills, understand their mistakes through human feedback, and track their progress over time. The system will simulate the question-and-answer flow of a real interview without needing a physical interviewer or a camera.

FACULTY SUPERVISOR REVIEW & AUDIT ASSESSMENT:

some changes need to there in your project .Too many modules, need to finalize the module. Need to study existing system to understand the exact flow of project.

Tanvi T. Joshi	Ms. Kalpana Dhende	AUTO	Head of Department
Lead Sign-off / Date	Internal Faculty Supervisor	External Industry Mentor	Dept. of Computer Applications (MCA)